POWER SERIES RINGS OVER FIELDS AND COHEN RINGS ARE FORMALLY SMOOTH OVER THE PRIME RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Universal property of power series rings. used at proof line 9719. Reduces the three adic statements to statements about the coefficient rings.
- Characterisation of formally smooth field extensions by separability. Algebra, Proposition \ref{algebra-proposition-characterize-separable-field-extensions}, cited at proof lines 9723-9724. Gives the characteristic zero and positive characteristic cases.
- Formal smoothness of a Cohen ring over the integers. Algebra, Lemma \ref{algebra-lemma-cohen-ring-formally-smooth}, cited at proof lines 9725-9726. Gives the third case, with the topological hypothesis checked on the test diagram.

Source reference. Original lemma 07NL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9766–9766. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0322: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4005–4005. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0329: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4012–4012. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07EB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9441–9441. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let K be a field of characteristic 0 and $A = K[[x_1, \ldots, x_n]]$. Let L be a field of characteristic $p > 0$ and $B = L[[x_1, \ldots, x_n]]$. Let Λ be a Cohen ring. Let $C = \Lambda[[x_1, \ldots, x_n]]$.*

(1) *$\mathbf{Q} \to A$ is formally smooth in the $\mathfrak{m}_A$-adic topology.*
(2) *$\mathbf{F}_p \to B$ is formally smooth in the $\mathfrak{m}_B$-adic topology.*
(3) *$\mathbf{Z} \to C$ is formally smooth in the $\mathfrak{m}_C$-adic topology.*

Proof. By the universal property of power series rings it suffices to prove:

(1) $\mathbf{Q} \to K$ is formally smooth.
(2) $\mathbf{F}_p \to L$ is formally smooth.
(3) $\mathbf{Z} \to \Lambda$ is formally smooth in the $\mathfrak{m}_\Lambda$-adic topology.

The first two are Algebra, Proposition 0322. The third follows from Algebra, Lemma 0329 since for any test diagram as in Definition 07EB some power of p will be zero in A/J and hence some power of p will be zero in A. □